Arjun Rao

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Education

Colby College 2026–2030

Prospective Math and Computer Science Major, 1/10 Pulver Science Scholars (providing 2 guaranteed research opportunities.) Research Assistant at Linde Packman Lab working on Computational Virology.

The Hyderabad Public School, Begumpet 2022–2026

98.4% ICSE (Highest Aggregate in School and AIR 8), A*A*A*A*AA in A Levels (Physics, Chemistry, Economics, Math, Further Math (self-studied) and English General Paper AS Level); 1560/1600 SAT Superscore (800 Math, 760 Reading and Writing); Class Valedictorian; National and Zonal Medals in Math and Science Talent Search; Deputy Headboy.

Professional Experience

Independent Research in AI Safety May 2026–Present

- Conducted independent mini-research projects in mechanistic interpretability, investigating the effects of induction head ablation on repetition dynamics in transformer language models using mechanistic techniques, and additionally worked on an AI Welfare mini-project analyzing How LLM ethical decision-making varies with Modifications to the Scaffold.
- Authored technical essays on AI safety and alignment, including Gödel's Incompleteness in AI, Do AI Chatbots Understand? and Beyond Alignment: Human Agency as a Design Objective for Frontier AI.
- Developed a metric for model valence based on activations in the J-space and causally validated it.

Founder, KalyanSarathi June 2024–June 2026

- Developed a bilingual AI chatbot, called KalyanSarathi, that advises users on government schemes they are eligible for and helps them apply.
- Improved awareness about schemes for 300+ people, granting them access to benefits worth INR 3 million.
- Programmed a fully TTS-STT enabled Android app that was recognized by the Ministry of Rural Development, Government of India. The project was open-sourced and handed off to another student maintainer.

Research Science Institute India (RSI-I), IISc Bengaluru, Jun 2025 – Jul 2026

- Selected as 1 of 31 students nationally by the Center for Excellence in Education.
- Conducted research in chemometrics at the Indian Institute of Science, Department of Biochemistry, under the Department Chair, Professor Utpal Tatu.
- Developed a novel machine learning and Raman spectroscopy method to detect the presence of *S. aureus* bacteria in milk samples.
- Received the award for Best Oral Presentation at the RSI Colloquium.
- Continued research after the summer to develop a *A Raman Consensus Spectra-based method for Milk Adulterant Detection* (co-first author paper) that will soon be submitted to a journal for publication.

Fellow, Global Public Policy and Leadership Fellowship Aug 2025–Aug 2025

- Selected for a fellowship with an 8% acceptance rate and underwent mentorship by senior employees at the World Bank and the Czech Foreign Ministry.
- Wrote a white paper titled "Evaluating the Current Asylum System for Climate Refugees and Recommendations to Create a Global Policy" that combined policy analysis with differential equations to model refugee flows.

Student Lead, Decarbonize Initiative Aug 2024–Nov 2024

- Selected to participate at COP29 in Baku, Azerbaijan, and granted access to the Blue Zone as a UNFCCC delegate.
- Presented the Global Child Manifesto in 2 panels at COP29 and personally pitched it to dignitaries, including the Canadian Climate Ambassador and the Director of IUCN.
- Conducted a study on the carbon sequestration capacity of The Hyderabad Public School, Begumpet, and presented the findings at the panels and the COP29 Science Fair.

Fernandez Hospital, Hyderabad May–Jul 2024

- First author of paper titled "Early Phototherapy Prediction for Neonatal Hyperbilirubinemia: A Machine and Deep Learning Based Approach," published *BMC Medical Informatics and Decision Making.*
- Built data scrapers to collect data, analyzed patient data, and built the ML model achieving 86% accuracy, a first of its kind for Indian neonates.
- Performed interpretability analysis, which led to the discovery of Cord TSH as a biomarker for severe hyperbilirubinemia.

Volunteer, People+AI and HaqDarshaq Aug 2024–Mar 2025

- Contributed to the mission statement and worked on DigiForm, a tool for automated form filling for illiterate and semi-literate individuals.
- Developed software for generating summaries of updates to government schemes for HaqDarshaq using LLM APIs. The software has been deployed for 660K+ users of the platform.

Editorial Intern, One Young India Apr–Jun 2022

- Wrote video scripts and conducted research for 2 videos on "Private Military Contractors" and "Oil Politics" that generated 500+ views.
- Authored a three-part Medium series on prediction in geopolitics and global economics. The articles were titled: "After London: The Search for the Next Global Kleptocratic Hub," "The Coming African Spring: Why the Sahel May Be Poised for Revolution," and "The Arctic: The Next Frontier in the US-China Rivalry."

Publications and Preprints

Rao, A., Oleti, T. P., Tanduri, A., Chandra, S., Sai Amarnath, V., Gummadi, P., Sreembhatla, N., & Sai Kiran, D. (2026). Early phototherapy prediction for neonatal hyperbilirubinemia: A machine and deep learning-based approach for Indian cohorts. BMC Medical Informatics and Decision Making. https://doi.org/10.1186/s12911-026-03839-w

Rao, Arjun. "Evaluating the Current Asylum System for Climate Refugees and Recommendations to Create a Global Policy." *International Journal of Applied Public Policy*, December 2025. https://www.ijapp.org/index.php/pub/article/view/5/2

Rao, Arjun. (2025). Applications of the Fourier Transform as a Financial Indicator for Cyclical Asset Prices. 10.2139/ssrn.6157848.

Rao, A. Interpretable Predictive Modeling of Host Specificity of Rabies lyssavirus from Glycoprotein Sequences. Research Square; 2026. DOI: 10.21203/rs.3.rs-9015851/v1.

Rao, Arjun. Why do Wars become so Difficult to End? Eliva Press, 2026. ISBN: 978-99993-5-677-0

Awards and Honors

- IRIS National Science Fair and ASTRYC – National Finalist (2025, 2026)
- John Locke Essay Competition (Economics) – Commendation and Invitation to Oxford Gala (2025) [Essay]
- Queen's Commonwealth Essay Competition – Silver Prize (2022)
- 20+ awards in Debate, MUN, and Theatre (including IPSC and ASISC) (2022-25)

Skills

Programming: Python, Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, TransformerLens
Technical: Statistical Modeling, Bioinformatics, Mathematical Modeling, AI/ML, Android App Development
Tools: GitHub, Data Pipelines, LLM Applications, RAG and Agentic Architectures
Other: Research Writing, Public Speaking

Interests

Cryptography, AI Safety, History, Computational and Mathematical Biology, Equestrian Sports.